

To: Dr. Pete Carson

From: CWC Gen, Christian Brown, Naomi Echo, Alonso Garcia, Javan Jake, Kaitlyn Redman

Date: 03/21/2025

Re: Analysis Memo

Number of Turns in a Coil

Naomi Echo will be using ANSYS to model how the number of turns in a coil affects the performance of the generator. To demonstrate this different number of turns will be modeled. These coils will be modeled in one of the preexisting generators. The number of turns already used in the generator will act as the base line, a lower and higher number of turns will be tested to demonstrate its impact.

Magnetic Flux and Generator Power in Ansys Motor CAD

Utilize the ANSYS Motor CAD to model in 2D. The goal is to find the magnetic flux and power generated with the matched current generator. Then modify the magnetic flux by changing the stator and rotor parameter. This will allow us to decide which specification to use for wire, efficiency, and optimize axial flux.

Wire Gauge in a Generator

Ensuring the correct wire gauge in a generator is crucial for maintaining a low Kv Rating. ANSYS provides a powerful way to test different wire gauges by simulating real-world conditions. To test wire gauge stability, Kaitlyn will run electromagnetic and thermal simulations to assess how different wire gauges impact current density, resistive losses, and temperature rise. The voltage drop across the winding will also be analyzed to ensure the wire gauge can handle the required load without excessive power loss. This approach will ensure that the generator's wiring is both reliable and optimized for real-world operation.

Arduino Code

Alonso will be working with an Arduino, acting like a data collector, that will collect data from sensors and implementations, like a voltage divider, current sensor, load amp, and infrared sensor. This is important for the success of this project because it'll help the team understand a generator's performance and lead to improvements on a generator. Also, this will allow the Arduino to be modular, as in it'll be able to implement more sensors and implementations, if needed. Lastly, it will be needed as part of our final steps when designing our initial generators.

Magnet Specifications

Christian Brown will be commencing alterations in magnet specifications to observe changes in the performance of the generator. This will be done in ANSYS Maxwell with 3D CADs and simulations of existing generators. Changes in dimensions, magnet pairs, and magnetic flux will be used to determine the most effective alterations that can be made to enhance performance. Doing this in a CAD software will allow quick changes in dimensions with instant results as opposed to altering the physical generators. Using the ANSYS software to acquire these results is one of the requirements given to us by our client.